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$$t^* = 6.62 \times 10^{-7} \exp\left(\frac{11949}{T}\right) = 6.62 \times 10^{-7} \exp\left(\frac{11949}{625}\right)$$

$$= 133 \text{ days}$$

$$t^* < t = 500 \text{ d}$$

$$\delta^* = 5.1 \exp\left(-\frac{550}{T}\right) = 5.1 \exp\left(-\frac{550}{625}\right) = 2.12$$

$$K_L = 7.48 \times 10^6 \exp\left(-\frac{12500}{625}\right) = 0.015$$

$$\Rightarrow f = \delta^* + K_L (t - t^*) = 2.12 + 0.015 (500 - 133)$$

$$= 7.625 \text{ } \mu\text{m}$$

$$(6) f = 0.14$$

$$f_{\text{ZrO}_2}^0 = \frac{32}{32+91} = 0.26$$

$$C_H^{\text{clad}} = \frac{2(0.14) \times (7.625) \times (5.68) \times (0.26) \times \left(\frac{1}{16}\right)}{\left(500 - \frac{7.625}{1.56}\right) \times (6.5)} \times 10^6$$

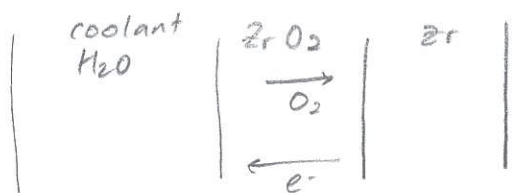
$$= 61.23 \text{ wt. ppm}$$

2. Oxidation is the rate-limiting step.

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-diffusion of oxygen

3. Hydrides form on the oxide layer of the cladding.



why? because

Hydrides form during this chemical reaction in ZrO₂.

-missed a bunch

Hydrides causes reduction in fracture toughness of cladding and causes embrittlement. ✓

4) H is also a design basis accident.

4. RIA is reactivity initiated accident, where a sudden insertion of reactivity occurs.

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Typical RIA in PWR is control rod system ejection due to system malfunction or if there is a break in the containment and rapid pressure decrease causes the control rod system to be ejected. In BWR, typical RIA is when the control blade is dropped suddenly, and stuck _{in} withdrawn position.

During an RIA, power increases rapidly and temperature also increases rapidly, which can cause PCMI and/or DNB Lased failure. -needed more

→ design basis accident.

5. LOCA stands for loss of coolant accident, which occurs when the coolant pipe breaks, and leads to SCRAM.

When a LOCA occurs, fuel is not cooled properly, which increases temperature and pressure of the fuel rod.

This can cause fuel melt or fuel rod ballooning / burst.

LOCA occurs at a much slower pace than KLA, which happens almost instantly ($0.1 s$)

6. High burnup leads to increased fission gas release, increased oxidation, increased hydrogen embrittlement, higher rod internal pressure, and PCMI.
 - need more

7. Improve reaction kinetics of steam, improve cladding properties, improve fuel properties, and enhance retainment of fission product.

8. Coating the cladding with ceramic, as this will increase resistance to oxidation and hydrides.
 - high T steam

9. High temperature steam increases oxidation layer and hydrides, which causes the cladding to become more brittle.
 - needed more

10. PCMI (Pellet Cladding Mechanical Interaction),
cladding wear, and
Departure from nucleate boiling.

11. CRUD hinders heat transport from fuel to coolant and
can block the flow way of coolant, which reduces the
fuel centerline margin. - one part

12. Addition of boric acid to control reactivity
and addition of LiOH to control the pH level, preferably > 6.9
to improve oxide & hydride formation.
Addition of Zinc to control radiation.
- how?

↳ In MOX fuel,

13. Bubbles migrate toward the center of the fuel pellet to form
a central void, due to evaporation and condensation mechanism
in the fuel, at $\sim 1800^\circ\text{C}$.

Below 1800°C , equiaxed grain growth is observed with larger
grains and at periphery, temperature is low so it shows
as-sintered grain structure.

- this is the same phenomenon, two parts